

Lesson 04(a)

Soil Classification



Topics for the day:

1. What is Soil classification
2. Soil classification by particle size
3. Particle size distribution tests
 - Sieve analysis test
 - Hydrometer test

What is Soil Classification ?

Soil classification is the separation of soil into classes or groups each having similar characteristics and similar behaviour.

A classification for **engineering purposes** should be based mainly on engineering properties of soil, eg. permeability, compressibility , strength.

Engineering behavior of soil primarily depend on **particle size.**

How are Soils Classified ;

Soils can be separated as coarse grained soils and fine grained soils.

Coarse grained soils are classified according to the particle size using two Particle Size Distribution Tests; sieve analysis test & hydrometer test.

Fine grained soils are classified according to the water contents at which they under go changes in consistency. Atterberg tests are used for this classification.

Particle Size Distribution Tests

Particle size distribution of a soil sample is measured using 02 tests.

- Sieve analysis test (sizes $> 0.075\text{mm}$)
- Hydrometer test (sizes $< 0.075\text{mm}$)

Full details of these tests are given in BS 1377: "Methods of test for soil for civil engineering purposes" and relevant ASTM standards.

Sieve Analysis Test

Sieve analysis test can be conducted in 02 ways;

Dry sieving

Soil sample (with fines removed) is dried and shaken through a nest of sieves placed in descending order.

Wet sieving

Separation of fine grains from coarse grains is done by washing the soil specimen.

Nest of Sieves



Procedure for Dry Sieving

- Weigh the oven dried soil sample
- Arrange the set of sieves in the descending order of size with a pan placed at the bottom
- Place the sample on the top sieve, cover with the lid and shake the set of sieves
- Mass of soil retained on each sieve is weighed

Observations

Sieve size (mm)	Mass retained (g)	% retained	Cumulative % retained	Cumulative % passing
4.75	A			
2.00	B			
0.850	C			
0.425	D			
0.150	E			
0.075	F			
pan	G			
total	W			

Calculations

- Calculate the % retained on each sieve
- Calculate the cumulative % retained on each sieve
- Calculate the cumulative % passing each sieve

Problem

A soil sample with a mass of 500g is placed in a nest of sieves ranging in size from 4.75mm to 0.075mm with a pan placed under the nest to catch all grains passing through the 0.075mm sieve. Observations are recorded in table 4 -1.

Complete the table and determine the particle size distribution of the soil sample.

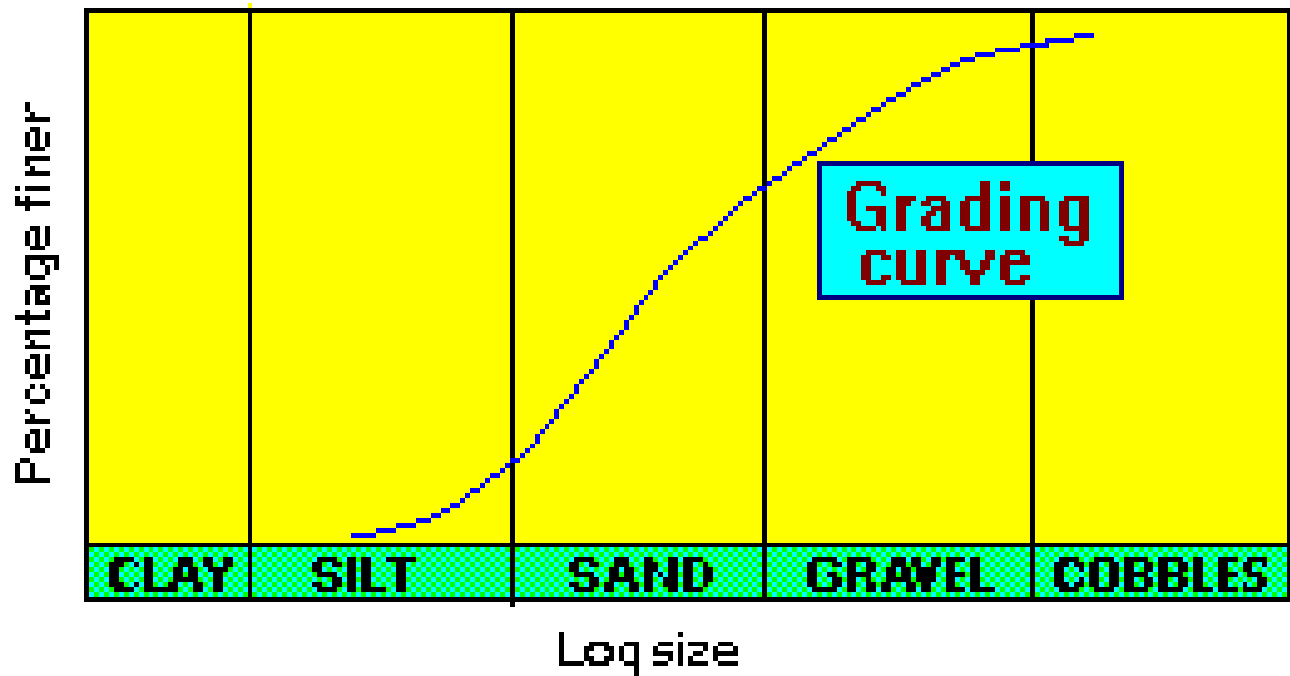
Table 4-1

Sieve size (mm)	Mass retained (g)	% retained	Cumulative % retained	Cumulative % passing
4.75	0	0%	0%	100%
2.00	50	10%	10%	90%
0.850	75	15%	25%	75%
0.425	175	35%	60%	40%
0.150	125	25%	85%	15%
0.075	50	10%	95%	5%
pan	25	5%	100%	0%
total	500			

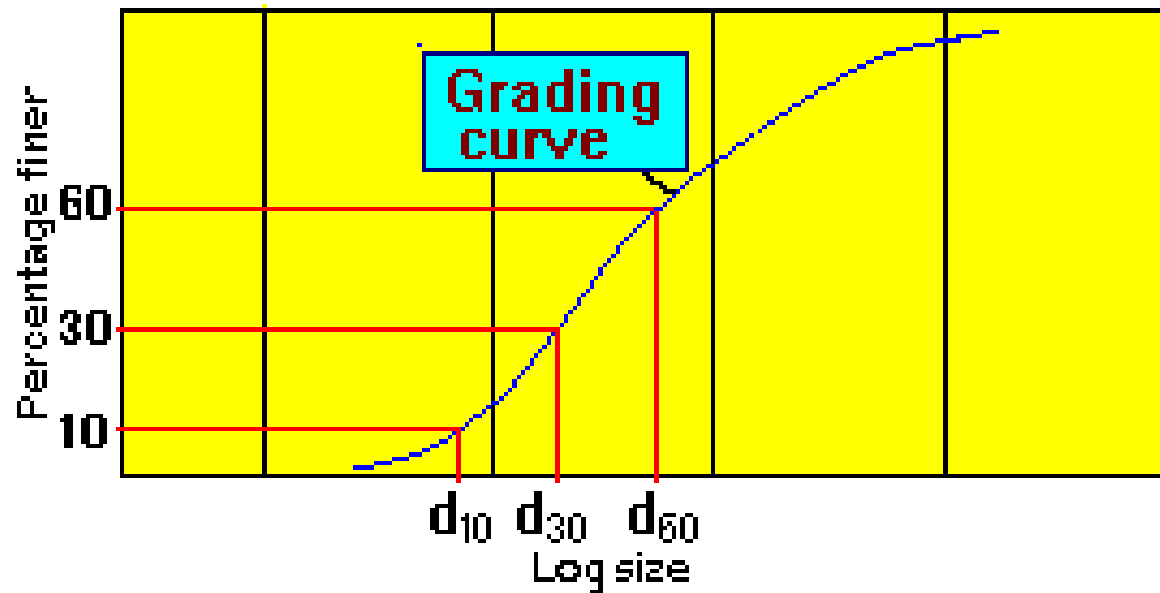
Results

Calculated results are then plotted on a % finer (passing) against log size graph.

A smooth S-shaped curve drawn through these points is called a **grading curve**.



Geometrical Grading Characteristics



three important sizes located on the horizontal axis are determined using the grading curve:

d_{10} = 10% of the particles are finer than this size

d_{30} = 30% of the particles are finer than this size

d_{60} = 60% of the particles are finer than this size

considering TABLE 4-1 ,

Grading Characteristics Calculated are:

Effective size

$$d_{10}$$

Uniformity coefficient

$$C_u = d_{60} / d_{10}$$

Coefficient of gradation

$$C_k = d_{30}^2 / d_{60} \times d_{10}$$

Coefficient of gradation is also called the Coefficient of curvature (C_c)

Classification of a Soil using the Grading Characteristics

$C_u > 5$ indicates a **well-graded soil**

$C_u < 3$ indicates a **poorly graded soil**

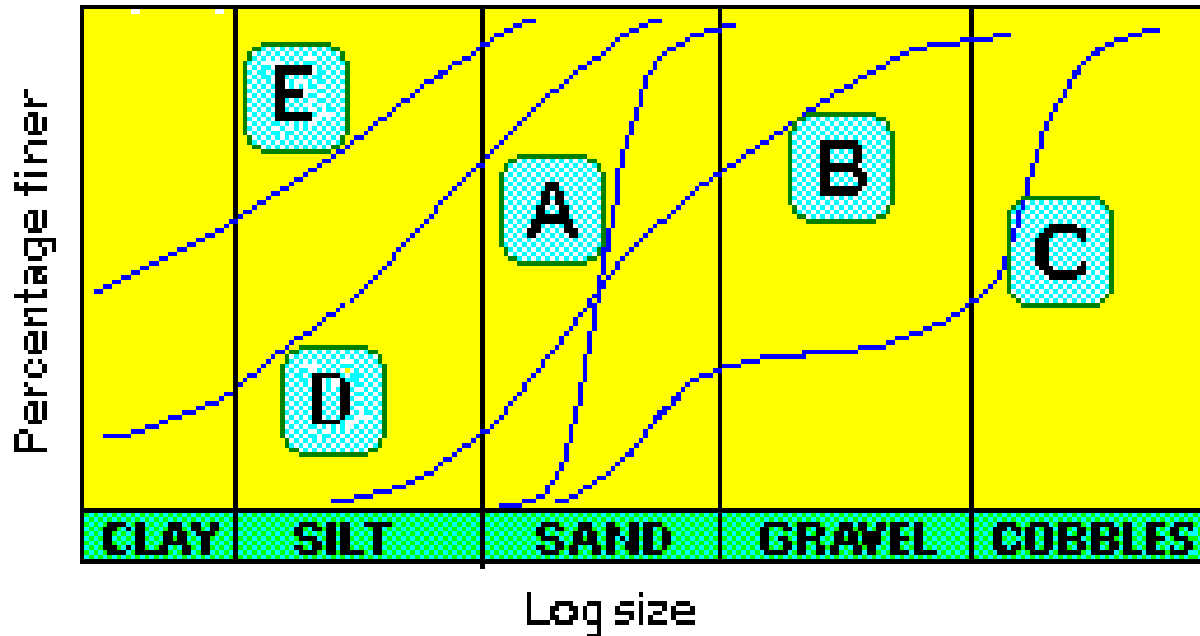
$0.5 < C_k < 2.0$ indicates a **well-graded soil**

$C_k < 0.1$ indicates a possible **gap-graded soil**

Both C_u and C_k will be 1 for a **single-sized soil**

Typical Grading Curves

The **position** and **shape** of the grading curve determines the soil class.



- A - a poorly-graded medium SAND
- B - a well-graded GRAVEL-SAND
- C - a gap-graded COBBLES-SAND
- D - a sandy SILT
- E - a typical silty CLAY

Hydrometer Test

Particles smaller than 0.075mm are too small for size determination by sieving.

Therefore a different technique - **sedimentation** - is used to determine the particle size distribution .

In the hydrometer analysis the soil sample is placed in a suspension and time is allowed to settle.

By use of **Stokes' Equation** the equivalent particle size of soil in suspension are computed.

Stokes Law governs the velocity at which spherical particles settle in a suspension.

The larger the particles the greater is the settling velocity.

Stokes Equation:

$$V = \left[\frac{\gamma_s - \gamma_w}{18\mu} \right] D^2$$

Where,

V = Terminal velocity of soil particle (cm/s)

D = Diameter of soil particle (cm)

γ_s = unit weight of soil particle

γ_w = unit weight of water

μ = viscosity of water (g-s/cm²)

The hydrometer measures the specific gravity of the soil suspension at the center of its bulb.

The specific gravity depends upon the mass of solids present in the suspension, which in turn depends upon the particle size.

The lower the specific gravity of the suspension the more the hydrometer will sink.



hydrometer

Procedure

- Soil sample is dispersed in water with a dispersing agent and allowed to settle
- Specific gravity of the suspension is measured at a depth ' L ' using a hydrometer at different times
- Using Stokes equation maximum particle size present in suspension at time ' t ' and depth ' L ' are calculated
- Results are plotted on a semi-log graph sheet

In many instances, the results of sieve analysis and hydrometer analysis for a given soil are combined on one graph.

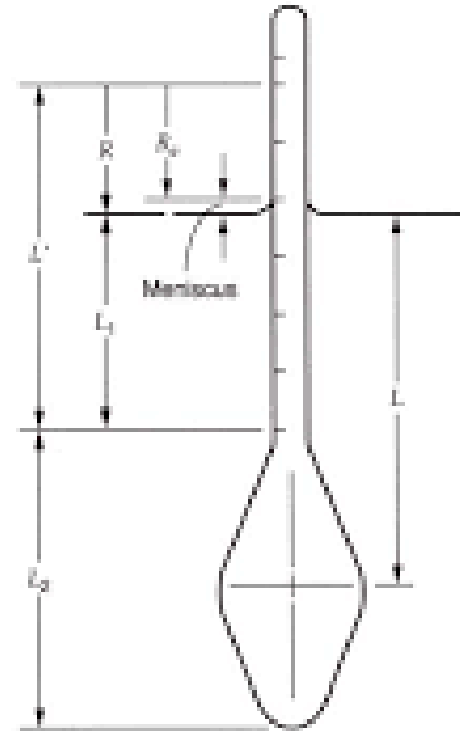
Calculations

Stokes Equation

$$V = \left[\frac{\gamma_s - \gamma_w}{18\mu} \right] D^2$$

From the diagram;

$$V = \frac{L}{t}$$



- Hydrometer reading observed will be related to the effective depth 'L'
- Corrected hydrometer reading will be related to the % finer
- % finer will be plotted against the particle size in a semi - log graph

Summary;

Engineering behavior of soils mainly depend on the particle size.

Soils are classified according to the particle size using the particle size distribution tests.

Sieve analysis test; size $> 0.075\text{mm}$

Hydrometer test; size $< 0.075\text{mm}$

Results of both tests are plotted on the same semi - log graph;
% finer(passing) vs particle size.

The class of the soil is determined by the position and the shape of the grading curve.